SoilPulse-ET: SmallSat Onboard Triage for Regenerative Agriculture Water-Stress Follow-Up

Evidence boundary: this Phase 1 artifact is a concept and policy demonstrator, not flight software or validated agronomic prediction.

Introduction

SoilPulse-ET is a SmallSat onboard-processing concept for regenerative agriculture and land-resilience monitoring. The system maintains a tile-level state ledger, then helps a spacecraft decide which candidate field observations should receive high-detail image chips, which should be compressed into compact feature summaries, and which should be deferred during constrained operations.

The first target user is an irrigation district, conservation planner, or regenerative agriculture technical-service provider supporting field teams under water-stress conditions. In a Central Valley-style pilot scenario, an analyst may need to prioritize cover-cropped orchards, deficit-irrigated vineyards, rotational pastures, riparian buffers, and managed fallow trials after a hot dry week. The operational challenge is not a lack of Earth-observation data. It is deciding which data are worth collecting, processing, and returning first under SmallSat power, compute, storage, and bandwidth limits.

SoilPulse-ET addresses the challenge as a systems problem. For creativity, it uses an onboard chip-versus-summary-versus-defer policy rather than a ground-only dashboard. For technical feasibility, it uses explicit resource budgets and degradation modes. For impact, it reports downlink saved, high-stress retention, and cloud-waste avoidance. For business model evaluation, it names institutional agricultural decision-support users and extension markets. For presentation, the paper, software artifact, and video evidence packet use the same dataflow and metrics.

Solution Architecture

The architecture begins with user priorities and field history on the ground. An observation planner converts those goals into candidate tile tasks. After acquisition, an onboard feature extractor produces a compact feature row for each tile: evapotranspiration anomaly proxy, vegetation-index change, cloud fraction, days since last useful observation, user priority, confidence, and resource estimates. A tile state ledger derives both a priority score and an evidence-quality score, so stale, cloudy, or low-confidence observations remain visible as operational knowledge. The triage policy then selects one of three packet actions.

Ground goal	Onboard processing	Packet decision	Ground use
User priorities and field history	Feature extraction: ET proxy, vegetation change, cloud, confidence	Triage policy selects chip, summary, or defer	ECOSTRESS-calibrated review and user worklist

Figure 1. Compact SoilPulse-ET architecture. Rendered diagram assets are provided separately.

Priority chips send higher-detail crops for immediate review. Feature summaries send compact evidence and metadata when the signal is useful but a full chip is not justified. Deferred tiles send nothing during the current contact, either because the tile is cloudy, below threshold, or does not fit available resources.

Check	If true	If false	Reason code
Cloud fraction high?	defer	score tile	cloud screen
Score below summary threshold?	defer	continue	below threshold
Score above chip threshold?	try priority chip	try summary	selected
Chip too large but summary fits?	feature summary	defer	summary kept or resource limit

Figure 2. Packet decision rules.

The current code artifact implements this policy as a transparent Python demo. Each tile receives a priority score, evidence score, knowledge-state label, action, reason code, and resource cost. The policy can downgrade a high-score chip to a summary if the chip does not fit the current budget.

NASA Data Anchor And Demo

The primary NASA data anchor is ECOSTRESS Tiled Evapotranspiration Instantaneous and Daytime L3 Global 70 m V002, DOI `10.5067/ECOSTRESS/ECO_L3T_JET.002`. ECOSTRESS supports the evapotranspiration logic and future calibration path. The implemented AppEEARS path produced a tiny ECOSTRESS-derived public micro-tile fixture without claiming validation on ECOSTRESS. The current submission-style demo uses a hybrid fixture: one ECOSTRESS-derived knowledge-gap row plus labeled synthetic mission-scenario rows for planned onboard behavior.

The demo maps ECOSTRESS and mission concepts to feature fields: ET estimates to `et_anomaly_mm_day`, uncertainty to `confidence`, cloud mask to `cloud_fraction`, water mask to excluded pixels, revisit history to `days_since_seen`, and user planning value to `user_priority`.

Tile	Action	Pri./evid.	State	KiB
covercrop	chip	0.74 / 0.77	clear_recent	96
vineyard	chip	0.71 / 0.73	stale	112
pasture	summary	0.55 / 0.78	clear_recent	8
orchard	summary	0.44 / 0.62	stale	8

riparian cloud	defer	0.32 / 0.51	obscured	0
riparian ECOSTRESS	defer	0.28 / 0.28	low_conf.	0
rangeland	defer	0.27 / 0.61	stale	0

Table 1. Hybrid demo packet decisions for real and synthetic mission rows.

Feasibility And Impact

The default demo budget is `384 KiB`, `6 packets`, `1500 ms`, and `42 J`, with planning envelopes of `128 MiB` memory and `512 MiB` temporary storage. The present-tense `60%` plan is anchored to a CDS Rev. `14.1` rail-based `6U` envelope (`100.0 x 226.3 x 366.0 mm`, typical max mass `12 kg`) without relying on optional extra volume, propulsion, or required deployables. The future `40%` path covers component selection, dispenser-specific refinement, and qualification work over the next `24` months. The representative compute stack is a radiation-tolerant CPU for control and policy execution, with optional FPGA or NPU acceleration for feature extraction.

Metric	Value
Full-chip baseline	784.0 KiB
Used downlink	224.0 KiB
Downlink saved	560.0 KiB / 71.43%
High-stress retention	2/2 / 100%
Cloudy candidates deferred	1/1
Cloud waste avoided	96.0 KiB
Processing budget used	48.00%
Energy budget used	38.10%

KiB	Pkts	Used	Saved	Stress
64	3	24.0	96.94%	100%
128	4	120.0	84.69%	100%
384	6	224.0	71.43%	100%

Table 2. Resource-use metrics and downlink sensitivity for the hybrid demo.

The smaller budgets force the policy toward compact summaries, which is the intended degraded behavior. The system does not make autonomous irrigation recommendations or guarantee yield outcomes. It creates a smaller, better prioritized analyst worklist.

Market, Execution, And Next Steps

The first adopter is an institutional user responsible for many fields: irrigation districts, conservation planners, regenerative agriculture program managers, or technical-service providers. NASA Harvest provides context for satellite-driven agricultural monitoring across policy and regional workflows. OpenET provides useful ET-market context for water budgets,

groundwater management, water trading, and ET-based irrigation practices, while reinforcing that ET data supports decisions rather than directly issuing irrigation commands.
Fractal Research Group builds public research infrastructure for targeted, evidence-bounded work, with current artifacts in longevity, biomedical evidence systems, and auditable research operations. SoilPulse-ET extends that operating style into land and water resilience, connecting FRG's broader mission around health, longevity, and fresh filtered water access for underserved communities to a NASA-grounded SmallSat concept. The team's role in Phase 1 is to make the system inspectable: reproducible code, explicit resource budgets, real ECOSTRESS lineage, and clear claim boundaries.
The MVP path is practical: expand the ECOSTRESS/AppEEARS event hunt across public windows and locations, calibrate thresholds with user feedback, extend the resource model to raw scene sizes and contact windows, and test degraded operations. Extension markets include forestry drought stress, restoration monitoring, post-fire recovery, and water-resource risk analytics.
Execution risk is controlled by keeping the Phase 1 artifact narrow: NASA data lineage, explicit resource accounting, transparent policy code, and clear separation between real observations and synthetic mission-scenario rows.
If selected as a finalist, the next steps are:

1. Expand the ECOSTRESS-derived event hunt across additional public date windows and locations.
2. Promote strong real stress-event rows into the hybrid fixture when evidence is clear enough to explain.
3. Calibrate thresholds with user priorities and field-scouting feedback.
4. Extend resource modeling to memory, storage, raw scene sizes, and contact windows.
5. Prepare a partner pilot with a district or conservation-service workflow.

Conclusion

SoilPulse-ET is a systems-first answer to the Space to Soil challenge. It uses a NASA evapotranspiration dataset as the scientific anchor, but centers the submission on onboard decision-making: what to sense, what to summarize, what to downlink, and what to defer under realistic SmallSat constraints. The code artifact gives judges a concrete view of the adaptive policy while preserving honest boundaries around flight readiness and agronomic validation.

References

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